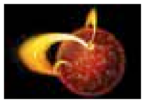
The big bang theory
The story of how pigeons helped scientists discover the cosmic microwave background radiation. **https://dgit.in/aug20-05**



Repeating FRB
Unravelling the 16 day cycle of a mysterious repeating FRB source by astronomers using CHIME. **https://dgit.in/aug20-06**

Mystery signals from space

Some of the observations by astronomical instruments just cannot be explained by conventional science

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In 1967, a pHD student at Cambridge, Jocelyn Bell, was studying radio signals from the sky in a telescope that was still being built. She found a rather odd repeating pattern with sharp pulses coming every 1.3 seconds from a particular region in the sky in the direction of the constellation of Vulpecula. Bell and her advisor, Anthony Hewish, began ruling out the possibilities that could produce the signal including other radio astronomers, radar probes bouncing off the moon, and interference from television signals, or orbiting satellites. When all these sources were ruled out, it was confirmed that the radio waves were indeed coming from a cosmic object. At that time, there was simply no known natural phenomenon that could cause such a periodic repeating signal, which made the scientists go towards the exciting possibility that they had finally stumbled upon signals from an intelligent civilization. For this reason, the object was designated LGM-1, or little green men. However, Bell found three more such sources in different parts of the sky and these could not all be intelligent extraterrestrials. Other astronomers joined in on the hunt and within a year dozens of pulsars were detected, which are rapidly rotating neutron stars. These objects were predicted to exist in 1933 but no one had actually found them. Walter Baade and Fritz Zwicky had previously predicted that after a star goes nova, it could produce a dense object made mostly of neutrons. Antony Hewish and another astronomer Martin Ryle who also built and developed radio telescopes were the first astronomers to be awarded the Nobel Prize in Physics, in 1974.

The basic story here is that a strange signals from an unknown source in space initially puzzles scientists, followed by a meticulous process of ruling out a number of sources, eventually leading to an exotic new source, confirming previous theoretical predictions. Karl Jansky's discovery of the first radio signals from a cosmic object turned out to be a supermassive blackhole in the centre of the galaxy, Sagittarius A*. Arno Penzias and Robert Woodrow Wilson had to eliminate all kinds of sources of interference on the discovery of cosmic microwave background radiation, including pigeon droppings on the telescope. Boyajian's star, which dimmed much more than transiting exoplanets should have allowed, was considered as a candidate for being a Dyson sphere, an artificial megastructure that harnesses most of the energies from a star. It is also called the "Where's The Flux?" star. Turns out that the dimming was caused by dust from excessive cometary activity. This is a story that repeats itself many